Emma's Atlas of Numbers and Shapes

- A not-so-brief guide to all the math I know -

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Sets

ALMOST all of modern mathematics can be put in the context of the study of ***sets*** and their ***elements***. Generally, given a collection of objects $a, b, c, d, \ldots$, they can be taken together as a set, generally written in curly brackets and represented by a capital letter:

$$M := \{a, b, c, d, e, \ldots\}$$

However, an approach where any collection of mathematical objects forms a set quickly runs into trouble - famously, ***Russel's paradox*** asks:

> *"Let R be the set of all sets that don't contain themselves. Does R contain itself?"*

If R doesn't contain itself, then by definition it must contain itself, but if it contains itself, then it cannot contain itself! If $R \in R$ is true, then it must be false! We arrive at a classic paradox. The conclusion, then, is that the set of all sets that don't contain themselves cannot exist - we cannot define sets to be any collection of things defined by any property, and must instead establish ground rules about which sets exist and exactly how we can use sets we already know to construct new ones.

These ground rules are given in the form of ***axioms***, which are statements that mathematicians take for granted, which can then be used as a starting point to then derive "the rest of mathematics" from. The following chapters contain a brief and hopefully easily understandable introduction to the ZFC axiom system, which was developed in the early 20th century by Ernst Zermelo and Abraham Fraenkel and is now the most widely accepted axiomatization of mathematics.

2.1 Infinity: $\emptyset$ and $\mathbb{N}$

2.2 Extensionality

To say "b is an element of M", or equivalently "b contains M", we simply write:

$$b \in M$$

One of the most basic axioms of set theory is the ***Axiom of Extensionality***, which states:

Axiom 2.2.1: Axiom of Extensionality

Two sets A and B are equal if and only if they contain the same elements.

The Axiom of Extensionality establishes that sets are uniquely defined by the objects they contain and have no additional structure. In particular:

1. The order of elements in a set doesn't matter. $\{1, 2\}$ and $\{2, 1\}$ are the same set.

2. The number of times an element appears within the same set doesn't matter. $\{1\}, \{1, 1\}$, and $\{1, 1, 1\}$ are the same set.

Definition 2.2.2: Subsets

We call a set A a ***subset*** of a set B if and only if every element of A is contained in B. We write "A is a subset of B" as $A \subseteq B$.

The demand that "every element of A is contained in B" can be stated more formally as "$x \in A$ implies $x \in B$". Importantly, this means that every set is a subset of itself - we always have $A \subseteq A$.

This definition gives us another way of phrasing the Axiom of Extensionality - two sets are equal if and only if each is a subset of the other:

$$A = B \Leftrightarrow A \subset B \text{ and } B \subset A$$

This idea is incredibly important for writing out proofs in practice - if a problem asks you to establish that two sets A and B are the same, then the easiest approach is almost always to show that any element of A must be contained in B, and to then show that any arbitrary element of B must be contained in A.

The subset relation is frequently written as $A \subset B$. However, other authors reserve $A \subset B$ to mean "A is a subset of B that isn't equal to B" - we call such a set a ***proper subset of B***. I will therefore attempt to only use the symbols $\subseteq$ and $\subsetneq$, which clear up this ambiguity.

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Chapter 1

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1.1 Arithmetic in n dimensions

We now focus our attention on n-dimensional euclidean space, which is modelled by the cartesian product $\mathbb{R}^n$:

$$\mathbb{R}^n = \left\{ \vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \;\middle|\; x_1, \ldots, x_n \in \mathbb{R} \right\} = \left\{ \vec{x} : \{1, \ldots, n\} \to \mathbb{R} \right\}$$

Elements of $\mathbb{R}^n$ are known as ***vectors***[1]. In the context of linear algebra, they are usually written vertically, as ***column vectors***.

Vectors simultaneously represent both ***positions*** and ***directions*** in space - in the same way that the number 5 is both the point that lies 5 units away from 0 on the real number line *and* the distance between the numbers 3 and 8 on that same number line, the vector $\binom{1}{2}$ represents both

- The point on the cartesian plane that can be reached by going one unit in the direction of one coordinate axis and then two units in the direction of the other axis, *and*
- the distance between two points with coordinates $\binom{2}{3}$ and $\binom{3}{5}$, or any other pair of points that are one unit apart in one coordinate direction and two in the other.

As is expected for algebraic structures defined on cartesian products, addition is defined **componentwise**:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

Given a real number c, we can scale elements of $\mathbb{R}^n$ by that number:

$$c \cdot \vec{x} = c \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} c \cdot x_1 \\ \vdots \\ c \cdot x_n \end{pmatrix}$$

[1] We will see in the next chapter that $\mathbb{R}^n$ is an example of a ***vector space***, including the case $n = 1$ (and $n = 0$, if you want to get a bit pathological). Therefore, formally speaking and only in certain contexts, regular old real (or even rational) numbers can be vectors too. However, when encountering the term "vector" in the wild, it is generally implied that $n \geq 2$.

The factor c is known as a ***scalar***, and the operator $\cdot : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ as ***scalar multiplication***. For the sake of readability, we differentiate scalars from vectors by writing a little arrow on top of a vector.

Arguably, it is these two operations - vector addition and scalar multiplication - that define linear algebra - next chapter, we will abstractly define ***vector spaces*** to be those structures where addition and scalar multiplication are well-behaved and the scalars come from a field, and ***modules*** to be their cousins whose scalars come from a ring.

In general, there is no well-behaved[2] vector multiplication that works for any n. Later on, we will prove this observation in the form of ***Frobenius' Theorem***, which roughly states:

[2] In this case, our criterion for multiplication being well-behaved is that multiplicative inverses should exist, i.e. that division should be possible.

Theorem 1.1.1: Frobenius' Theorem

The only dimensions in which an associative multiplication exists on $\mathbb{R}^n$ are $\mathbb{R}^1$, i.e. $\mathbb{R}$ itself, $\mathbb{R}^2$, where multiplication is given by treating two-dimensional vectors as complex numbers, and $\mathbb{R}^4$, where multiplication is given treating four-dimensional vectors as ***quaternions***.

We will take a closer look at the complex numbers $\mathbb{C} \cong \mathbb{R}^2$ and the quaternions $\mathbb{H} \cong \mathbb{R}^4$ in the following sections. In particular, we will see that quaternion multiplication isn't commutative, so if we require multiplication to be commutative, our options shrink even further, leaving just $\mathbb{R}$ and $\mathbb{C} \cong \mathbb{R}^2$.

1.2 The Inner Product: Angles and Lengths in $\mathbb{R}^n$

Another operation that is of central importance to euclidean geometry is the ***inner product***, also known as the ***dot product*** or ***scalar product***.

Definition 1.2.1: Euclidean Inner Product

Given two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$, their ***Euclidean inner product***, written $\left\langle \vec{x}, \vec{y} \right\rangle$ or simply $\vec{x} \cdot \vec{y}$, is the sum of the products of the components of $\vec{x}$ with the components of $\vec{y}$:

$$\left\langle \vec{x}, \vec{y} \right\rangle = \sum_{i=1}^{n} x_i \cdot y_i$$

Theorem 1.2.2: Properties of Inner Products

The Euclidean inner product has the following properties:

1. ***(Positive definiteness):*** For every $\vec{x} \in \mathbb{R}^n$, we have $\left\langle \vec{x}, \vec{x} \right\rangle = 0$ if and only if $\vec{x} = \vec{0}$.
2. ***(Linearity in the first argument):*** For every $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$, we have:

$$\langle ax + by, z \rangle = a \langle x, z \rangle + b \langle y, z \rangle$$

3. ***(Symmetry):*** For every $\vec{x}, \vec{y} \in \mathbb{R}^n$, we have:

$$\langle x, y \rangle = \langle y, x \rangle$$

Exercise 1.2.3. *Show that the Euclidean inner product fulfills the following version of the binomial formula:*

$$\left\langle \vec{x} + \vec{y}, \vec{x} + \vec{y} \right\rangle = \left\langle \vec{x}, \vec{x} \right\rangle + 2 \left\langle \vec{x}, \vec{y} \right\rangle + \left\langle \vec{y}, \vec{y} \right\rangle$$
$$\left\langle \vec{x} - \vec{y}, \vec{x} - \vec{y} \right\rangle = \left\langle \vec{x}, \vec{x} \right\rangle - 2 \left\langle \vec{x}, \vec{y} \right\rangle + \left\langle \vec{y}, \vec{y} \right\rangle$$

The importance of inner products comes from the fact they let us define ***lengths*** and ***angles***.

Definition 1.2.4: Euclidean Norm

The length of a vector $\vec{x} \in \mathbb{R}^n$, also known as its ***Euclidean norm***, written as $\|\vec{x}\|_2$, $\|\vec{x}\|$, or just $|x|$, is given by:

$$\|\vec{x}\|_2 = \sqrt{\left\langle \vec{x}, \vec{x} \right\rangle} = \sqrt{x_1^2 + \ldots + x_n^2}$$

Note that for $n = 1$, we simply get $\|x\|_2 = \sqrt{x^2} = |x|$, so any result concerning the Euclidean norm on $\mathbb{R}^n$ also holds for the absolute value function.

Theorem 1.2.5: Some properties of Norms

The Euclidean norm has the following properties:

1. ***(Non-negativity):*** For every $\vec{x} \in \mathbb{R}^n$, $\|\vec{x}\|_2 \geq 0$.
2. ***(Positive definiteness):*** For every $\vec{x} \in \mathbb{R}^n$, $\|\vec{x}\|_2 = 0$ if and only if $\vec{x} = \vec{0}$.
3. ***(Absolute homogeneity):*** For every $c \in \mathbb{R}$ and $\vec{x} \in \mathbb{R}^n$, we have
$$\|c \cdot \vec{x}\|_2 = |c| \cdot \|\vec{x}\|_2$$

The little subscript 2 in $\|\vec{x}\|_2$ differentiates the Euclidean norm from the many other norms we will meet over time - in general, for any $p \in \mathbb{N}_1$, $\|\vec{x}\|_p$ denotes the function

$$\|\vec{x}\|_p = \sqrt[p]{|x_1|^p + \ldots + |x_n|^p} \qquad \left(= \sqrt[p]{\sum_{i=1}^{n} |x_i|^p}\right),$$

known as the ***p*-norm**. For now, we only really care about the Euclidean norm, but other values of p will start coming in handy once we start applying tools from analysis to vector spaces. The limiting case $p \to \infty$ is particularly useful: We will find that

$$\|\vec{x}\|_\infty = \max |x_i|.$$

Theorem 1.2.6: Cauchy-Schwartz Inequality

Let $\vec{x}, \vec{y} \in \mathbb{R}^n$. Then we have

$$|\langle \vec{x}, \vec{y} \rangle| \leq \|\vec{x}\|_2 \cdot \|\vec{y}\|_2$$

Proof.

1. Assume $\|\vec{x}\|_2 = 0$. Then we immediately get
$$\langle \vec{x}, \vec{y} \rangle = 0 = \|\vec{x}\|_2 \cdot \|\vec{y}\|_2.$$
2. Assume $\|\vec{x}\|_2 = \|\vec{y}\|_2 = 1$. Then we have $\|\vec{x}\|_2^2 + \|\vec{y}\|_2^2 = 2$, and thus:
$$\begin{aligned} 1 + \langle \vec{x}, \vec{y} \rangle &= \frac{1}{2}(\|\vec{x}\|_2^2 + \|\vec{y}\|_2^2) + \langle \vec{x}, \vec{y} \rangle \\ &= \frac{1}{2}(\|\vec{x}\|_2^2 + 2\langle \vec{x}, \vec{y} \rangle + \|\vec{y}\|_2^2) \\ &= \frac{1}{2}\|\vec{x} + \vec{y}\|_2^2 \\ &\geq 0 \end{aligned}$$
Thus we get $\langle \vec{x}, \vec{y} \rangle \geq -1$. Applying the same reasoning to $1 - \langle \vec{x}, \vec{y} \rangle$, we get
$$1 - \langle \vec{x}, \vec{y} \rangle = \frac{1}{2}\|\vec{x} - \vec{y}\|_2^2 \geq 0,$$
i.e. $\langle \vec{x}, \vec{y} \rangle \leq 1$. We thus have
$$|\langle \vec{x}, \vec{y} \rangle| \leq 1 = \|\vec{x}\|_2 \cdot \|\vec{y}\|_2,$$

as desired.

3. Now, let $\vec{x}, \vec{y} \in \mathbb{R}^n \setminus \left\{\vec{0}\right\}$ be arbitrary. Then $\frac{\vec{x}}{\|\vec{x}\|_2}$ and $\frac{\vec{y}}{\|\vec{y}\|_2}$ both have norm 1, letting us simply use our previous result:

$$\begin{aligned}
\left|\langle \vec{x}, \vec{y} \rangle\right| &= \left| \|\vec{x}\| \cdot \|\vec{y}\| \cdot \left\langle \frac{\vec{x}}{\|\vec{x}\|}, \frac{\vec{y}}{\|\vec{y}\|} \right\rangle \right| \\
&\leq \left| \|\vec{x}\| \cdot \|\vec{y}\| \cdot \left\| \frac{\vec{x}}{\|\vec{x}\|} \right\| \cdot \left\| \frac{\vec{y}}{\|\vec{y}\|} \right\| \right| \\
&= \|\vec{x}\| \cdot \|\vec{y}\|
\end{aligned}$$

□

Corollary 1.2.7: Triangle Inequality

For every $\vec{x}, \vec{y} \in \mathbb{R}^n$, we have

$$\|\vec{x} + \vec{y}\|_2 \leq \|\vec{x}\|_2 + \|\vec{y}\|_2$$

We had previously formalized the concept of an 'angle between two lines' via the complex exponential function, and then defined the trigonometric functions $\sin x$ and $\cos x$ as the complex and real parts of e^{ix}, respectively. If we take the cosine function as a given, then scalar products provide a convenient alternate way of defining angles, which works in any dimension:

Definition 1.2.8: Inner angle between two vectors

Given two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n \setminus \left\{\vec{0}\right\}$, we define the ***inner angle*** between them to be:

$$\measuredangle(\vec{x}, \vec{y}) = \arccos\left(\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|_2 \cdot \|\vec{y}\|_2}\right)$$

Note that Cauchy-Schwartz gives us $\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|_2 \cdot \|\vec{y}\|_2} \in [-1, 1]$, showing that this expression actually is well-defined.

Exercise 1.2.9. *Verify that we have:*

1. $\measuredangle(\vec{x}, \vec{y}) = 0$ *if and only if there exists* $c \in \mathbb{R}_{>0}$ *such that* $\vec{x} = c\vec{y}$*.*

2. $\measuredangle(\vec{x}, \vec{y}) = \frac{\pi}{2}$ *if and only if* $\langle x, y \rangle = 0$*.*

3. $\measuredangle(\vec{x}, \vec{y}) = \pi$ *if and only if there exists* $c \in \mathbb{R}_{>0}$ *such that* $\vec{x} = -c\vec{y}$*.*

Notably, this suggests that our definition is sensible even in the case $n = 1$, since any pair of non-zero vectors fulfills either condition 1 or 2, implying that the angle between two vectors on the real number line is either 0 or π.

These inner angles match our earlier definition of angles in terms of complex numbers, as long as we are content with not being able to meaningfully talk about negative angles or angles greater than π anymore:

Exercise 1.2.10. *Earlier in this book, we defined angles in terms of complex numbers. Given a complex number $y = re^{i\varphi}$ we saw that the corresponding real vector is given by $\vec{y} = \binom{r\cos\varphi}{r\sin\varphi}$.*

Show that our two definitions match as long as $\varphi \in [0, \pi]$, i.e. that given a vector $\vec{x} = \binom{x}{0}$ lying on the x-axis, we have

$$\measuredangle(\vec{x}, \vec{y}) = \varphi$$

(Hint: Remember $\cos^2\varphi + \sin^2\varphi = 1$)

We have to be content with this restricted interval $[0, \pi]$, since the cosine function isn't injective on larger intervals, meaning the arccosine would no longer be well-defined.

More generally, any definition of an 'angle between two arbitrary vectors' will suffer from this - if we only specify two vectors, it's left ambiguous which side we should measure the angle on.

This problem can be solved by finding a way to take a tuple $(\vec{x}_1, \ldots, \vec{x}_n)$ of vectors and recognize whether it is 'positively ordered' or 'negatively ordered' - in $\mathbb{R}^2$, this would be asking whether the second vector lies 'clockwise' or 'counterclockwise' from the first, while in $\mathbb{R}^3$ the question is whether the vectors are 'left-handed' or 'right-handed'. A map that gives us this information is known as an ***orientation***. We will formalize this idea later, when we have introduced the necessary tools, those being vector spaces, their ***bases***, and ***determinant functions***.

In the complex case, we could dodge this issue because one of our vectors, call it $\vec{x}$, was always part of the x-axis, and thus we could obtain a canonical orientation by declaring that the angle between $\vec{x}$ and $\vec{y}$ was positive if $y_2 > 0$ and negative if $y_2 < 0$.

Exercise 1.2.11. *Prove the following theorems of classical Euclidean geometry:*

1. *The **Law of Cosines**:*

$$\|\vec{x} - \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|^2 - 2\|\vec{x}\|_2\|\vec{y}\|_2 \cdot \cos(\measuredangle(\vec{x}, \vec{y}))$$

2. *The **Pythagorean Theorem**:*

$$\measuredangle(\vec{x}, \vec{y}) = \frac{\pi}{2} \implies \|\vec{x} - \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|^2$$

1.3 The Cross Product

The ***cross product*** is an operation $\times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$, such that:

1. $\vec{u} \times \vec{v}$ is orthogonal to both $\vec{u}$ and $\vec{v}$,
2. the length of $\vec{u} \times \vec{v}$ is the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$:

$$\left\|\vec{u} \times \vec{v}\right\|_2 = \left\|\vec{u}\right\|_2 \left\|\vec{v}\right\|_2 \sin\left(\measuredangle(\vec{u}, \vec{v})\right)$$

We also declare that our cross product should be ***anticommutative:*** given two vectors, we want to be able to change the sign of the cross product by switching the order in which we multiply:

$$\vec{v} \times \vec{w} := -\vec{w} \times \vec{v}$$

Notice how this immediately implies

$$\begin{aligned} & \vec{v} \times \vec{v} = -(\vec{v} \times \vec{v}) \\ \implies & \vec{v} \times \vec{v} = 0 \end{aligned}$$

Finally, notice that our geometric intuition immediately tells us that our product should be ***linear*** and ***distributive over vector addition***.

We can now derive an explicit formula for this operation on our own. First, note that the standard basis

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \vec{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

consists of orthogonal vectors of length one, so we can start by declaring

$$\begin{aligned} \vec{e}_1 \times \vec{e}_2 &:= \vec{e}_3 \\ \vec{e}_2 \times \vec{e}_3 &:= \vec{e}_1 \\ \vec{e}_3 \times \vec{e}_1 &:= \vec{e}_2 \end{aligned}$$

Note that once again, the direction in which these vectors point depends on a somewhat arbitrary choice of orientation - we could invert all of the directions and still end up with a valid definition.

These observations are already enough to calculate the cross product of two arbitrary vectors: Using distributivity, we get:

$$
\begin{aligned}
\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} &= (v_1\vec{e}_1 + v_2\vec{e}_2 + v_3\vec{e}_3) \times (w_1\vec{e}_1 + w_2\vec{e}_2 + w_3\vec{e}_3) \\
&= v_1w_1(\vec{e}_1 \times \vec{e}_1) \\
&+ v_1w_2(\vec{e}_1 \times \vec{e}_2) \\
&+ v_1w_3(\vec{e}_1 \times \vec{e}_3) \\
\\
&+ v_2w_1(\vec{e}_2 \times \vec{e}_1) \\
&+ v_2w_2(\vec{e}_2 \times \vec{e}_2) \\
&+ v_2w_3(\vec{e}_2 \times \vec{e}_3) \\
\\
&+ v_3w_1(\vec{e}_3 \times \vec{e}_1) \\
&+ v_3w_2(\vec{e}_3 \times \vec{e}_2) \\
&+ v_3w_3(\vec{e}_3 \times \vec{e}_3)
\end{aligned}
$$

To which we can apply anticommutativity and our formulas for multiplying the basis vectors to get:

$$
\begin{aligned}
&\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} \\
\\
&= v_1w_1 \cdot \vec{0} \\
&+ v_1w_2 \cdot \vec{e}_3 \\
&+ v_1w_3 \cdot (-\vec{e}_2) \\
\\
&+ v_2w_1 \cdot (-\vec{e}_3) \\
&+ v_2w_2 \cdot \vec{0} \\
&+ v_2w_3 \cdot \vec{e}_1 \\
\\
&+ v_3w_1 \cdot \vec{e}_2 \\
&+ v_3w_2 \cdot (-\vec{e}_1) \\
&+ v_3w_3 \cdot \vec{0} \\
\\
&= \begin{pmatrix} v_2w_3 - v_3w_2 \\ v_3w_1 - w_1v_3 \\ v_1w_2 - v_2w_1 \end{pmatrix}
\end{aligned}
$$

Corollary 1.3.1: Cross Product

The cross product $\vec{v} \times \vec{w}$ of two vectors $\vec{v}, \vec{w} \in \mathbb{R}^3$ returns a vector whose direction is orthogonal to both $\vec{v}$ and $\vec{w}$ and whose length equals the area of the parallelogral spanned between them. Given coordinate representations of $\vec{v}$ and $\vec{w}$, the cross product can be calculated via the following formula:

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - w_1 v_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}$$

Proposition 1.3.2. *We have*

$$\left\langle \vec{u} \times \vec{v}, \vec{w} \times \vec{t} \right\rangle = \langle \vec{u}, \vec{w} \rangle \left\langle \vec{v}, \vec{t} \right\rangle - \left\langle \vec{u}, \vec{t} \right\rangle \langle \vec{v}, \vec{w} \rangle .$$

Proposition 1.3.3. *We have*

$$\langle \vec{u} \times \vec{v}, \vec{w} \rangle = \langle \vec{v} \times \vec{w}, \vec{u} \rangle = \langle \vec{w} \times \vec{u}, \vec{v} \rangle$$

This expression is known as the ***box product*** *of $\vec{u}, \vec{v}, \vec{w}$. Its absolute value corresponds to the volume of the parallelepiped spanned by the three vectors.*

Proposition 1.3.4. *We have*

$$(\vec{u} \times \vec{v}) \times \vec{w} = \langle u, w \rangle \cdot v - \langle v, w \rangle \cdot u = w \times (v \times u).$$

This result is known as ***Grassmann's Identity****.*

Proposition 1.3.5. *We have*

$$(\vec{u} \times \vec{v}) \times \vec{w} + (\vec{v} \times \vec{w}) \times \vec{u} + (\vec{w} \times \vec{u}) \times \vec{v} = 0$$

This result is known as ***Jacobi's Identity.***

1.4 The Quaternions $\mathbb{H}$

Definition 1.4.1: Quaternions

The Quaternions $\mathbb{H}$ consist of the set $\mathbb{R}^4$, with the following special notation:

$$\begin{aligned} \vec{1} &:= (1,0,0,0) \\ i &:= (0,1,0,0) \\ j &:= (0,0,1,0) \\ k &:= (0,0,0,1) \end{aligned}$$

Such that in addition to scalar multiplication and vector addition, which work as usual, we have an additional vector multiplication operation $\cdot : \mathbb{R}^4 \times \mathbb{R}^4 \to \mathbb{R}^4$ that behaves as follows:

1. $\vec{1}$ is a neutral element of vector multiplication:

$$\begin{aligned} &\forall x \in \mathbb{H} : \\ &\vec{1} \cdot x = x \cdot \vec{1} = x \end{aligned}$$

2. Vector multiplication is associative:

$$\vec{x}(\vec{y}\vec{z}) = (\vec{x}\vec{y})\vec{z}$$

3. Multiplicative inverses exist:

$$\begin{aligned} &\forall \vec{x} \in \mathbb{H}_{\neq 0} : \\ &\exists \vec{x}^{-1} \in \mathbb{H} : \\ &\vec{x} \cdot \vec{x}^{-1} = \vec{x}^{-1} \cdot \vec{x} = \vec{1} \end{aligned}$$

4. Vector multiplication distributes over vector addition, from both sides:

$$\begin{aligned} &\forall \vec{x}, \vec{y}, \vec{z} \in \mathbb{H} : \\ &\vec{x} \cdot (\vec{y} + \vec{z}) = \vec{x}\vec{y} + \vec{x}\vec{z} \\ &(\vec{x} + \vec{y}) \cdot \vec{z} = \vec{x}\vec{z} + \vec{y}\vec{z} \end{aligned}$$

5. Vector multiplication is compatible with scalar multiplication:

$$\begin{aligned} &\forall a, b \in \mathbb{R}, \vec{x}, \vec{y} \in \mathbb{H} : \\ &a\vec{x} \cdot b\vec{y} = (ab)(\vec{x}\vec{y}) \end{aligned}$$

6. The unit vectors fulfill the ***Quaternion multiplication formula***:

$$i^2 = j^2 = k^2 = ijk = -\vec{1}$$

Apart from the Quaternion multiplication formula, which is the defining feature of the Quaternions, all of our other demands together state that $\mathbb{H}$ should be an ***associative real division algebra over*** $\mathbb{R}^4$.

One very important consequence of this definition is that vectors of the form $\vec{a} := (a, 0, 0, 0)$ behave exactly like real numbers - let $\vec{x} \in \mathbb{H}$, then we have:

$$\vec{a}\vec{x} = (a \cdot \vec{1}) \cdot \vec{x} = a \cdot (\vec{1} \cdot \vec{x}) = a\vec{x}$$

For this reason, it is common to write $\vec{1}$ simply as 1, and to then proceed as we did for the complex numbers and write:

$$(a, b, c, d) = a + bi + cj + dk$$

We can now already get most of the way to deriving multiplication formula for quaternions, by simply applying distributivity:

$$\begin{aligned}
&(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k) \\
&\qquad = v_1 u_1 + (v_1 u_2)i + (v_1 u_3)j + (v_1 u_4)k \\
&\qquad + (v_2 u_1)i + (v_2 u_2)i^2 + (v_2 u_3)ij + (v_2 u_4)ik \\
&\qquad + (v_3 u_1)j + (v_3 u_2)ji + (v_3 u_3)j^2 + (v_3 u_4)jk \\
&\qquad + (v_4 u_1)k + (v_4 u_2)ki + (v_4 u_3)kj + (v_4 u_4)k^2
\end{aligned}$$

Now, using $i^2 + j^2 + k^2 = ijk = -1$, we can simplify slightly:

$$\begin{aligned}
&(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k) \\
&\qquad = v_1 u_1 + (v_1 u_2)i + (v_1 u_3)j + (v_1 u_4)k \\
&\qquad + (v_2 u_1)i + (v_2 u_2)i^2 + (v_2 u_3)ij + (v_2 u_4)ik \\
&\qquad + (v_3 u_1)j + (v_3 u_2)ji + (v_3 u_3)j^2 + (v_3 u_4)jk \\
&\qquad + (v_4 u_1)k + (v_4 u_2)ki + (v_4 u_3)kj + (v_4 u_4)k^2 \\
&\qquad = v_1 u_1 + (v_1 u_2)i + (v_1 u_3)j + (v_1 u_4)k \\
&\qquad + (v_2 u_1)i - (v_2 u_2) + (v_2 u_3)ij + (v_2 u_4)ik \\
&\qquad + (v_3 u_1)j + (v_3 u_2)ji - (v_3 u_3) + (v_3 u_4)jk \\
&\qquad + (v_4 u_1)k + (v_4 u_2)ki + (v_4 u_3)kj - (v_4 u_4)
\end{aligned}$$

All that's left is to figure out what the products of i, j and k should be.

We can derive ij and ji as follows:

1. Start with $ijk = -1$.
2. Multiply k from the right to get $ijk^2 = -k$, i.e. $-ij = -k$, i.e. $ij = k$.
3. Multiply ij from the left to get $ijij = ijk = -1$.
4. Multiply i from the left and j from the right to get $i(ijij)j = -ij$.
5. Simplify the left side to get $i(ijij)j = (ii)(ji)(jj) = ji$, i.e:

$$ij = -ji = k$$

Exercise 1.4.2. *Show that we also have $jk = -kj = i$ and $ki = -ik = j$.*

This means that Quaternion multiplication isn't commutative!

Our final multiplication table for $1, i, j$ and k is thus:

$\cdot$	1	i	j	k
1	1	i	j	k
i	i	-1	k	$-j$
j	j	$-k$	-1	i
k	k	j	$-i$	-1

Exercise 1.4.3. *Verify that this multiplication table doesn't violate associativity.*

Entering these new results into our partial multiplication formula, we get:

$$\begin{aligned}
(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k)& \\
&= v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k \\
&+ (v_2 u_1) i - (v_2 u_2) + (v_2 u_3) ij + (v_2 u_4) ik \\
&+ (v_3 u_1) j + (v_3 u_2) ji - (v_3 u_3) + (v_3 u_4) jk \\
&+ (v_4 u_1) k + (v_4 u_2) ki + (v_4 u_3) kj - (v_4 u_4) \\
&= v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k \\
&+ (v_2 u_1) i - (v_2 u_2) + (v_2 u_3) k - (v_2 u_4) j \\
&+ (v_3 u_1) j - (v_3 u_2) k - (v_3 u_3) + (v_3 u_4) i \\
&+ (v_4 u_1) k + (v_4 u_2) j - (v_4 u_3) i - (v_4 u_4)
\end{aligned}$$

which we can finally simplify to get:

$$\begin{aligned}
(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k)& \\
= \ &(v_1 u_1 - v_2 u_2 - v_3 u_3 - v_4 u_4) \\
+ \ &(v_1 u_2 + v_2 u_1 + v_3 u_4 - v_4 u_3) i \\
+ \ &(v_1 u_3 - v_2 u_4 + v_3 u_1 + v_4 u_2) j \\
+ \ &(v_1 u_4 + v_2 u_3 - v_3 u_2 + v_4 u_1) k
\end{aligned}$$

Now, this is obviously a long and unreasonably complicated formula - is there any way to think of it differently? The complex numbers had a very nice and intuitive geometric definition, maybe the quaternions do too?

Well, the key trick here is to split the underlying set $\mathbb{R}^4$ into $\mathbb{R} \times \mathbb{R}^3$, thinking of each Quaternion as consisting of a ***real part*** $a \in \mathbb{R}$ and an ***imaginary part*** $\vec{v} \in \mathbb{R}^3$. Our multiplication formula now reads:

$$\begin{aligned}
&(a, \vec{v}) \cdot (b, \vec{u}) \\
= \ &(ab - v_1 u_1 - v_2 u_2 - v_3 u_3) \\
+ \ &(a u_1 + b v_1 + v_2 u_3 - v_3 u_2) i \\
+ \ &(a u_2 - v_1 u_3 + b v_2 + v_3 u_1) j \\
+ \ &(a u_3 + v_1 u_2 - v_2 u_1 + b v_3) k
\end{aligned}$$

which you might notice consists exclusively of familiar vector operations:

1. $ab - v_1u_1 - v_2u_2 - v_3u_3 = ab - \left\langle \vec{v}, \vec{u} \right\rangle$

2. $\begin{pmatrix} au_1 \\ au_2 \\ au_3 \end{pmatrix} = a\vec{u}, \begin{pmatrix} bv_1 \\ bv_2 \\ bv_3 \end{pmatrix} = b\vec{v}$

3. $\begin{pmatrix} v_2u_3 - v_3u_2 \\ v_1u_3 - v_3u_1 \\ v_1u_2 - v_2u_1 \end{pmatrix} = \vec{v} \times \vec{u}$

Putting everything together, we can now write out our multiplication formula in a much nicer form:

$$(a, \vec{v}) \cdot (b, \vec{u}) = (ab - \left\langle \vec{v}, \vec{u} \right\rangle, a\vec{u} + b\vec{v} + \vec{v} \times \vec{u}),$$

Exercise 1.4.4. *Verify that quaternion multiplication is associative and distributes over vector addition.*

Exercise 1.4.5. *Let $p = (a, \vec{v}) \in \mathbb{H}$. Then $pq = qp$ holds for all $q \in \mathbb{H}$ if and only if $\vec{v} \neq 0$.*

Similar to complex numbers, we can define ***quaternion conjugation***:

Definition 1.4.6: Quaternion Conjugation

Just like for $\mathbb{C}$, we define the ***conjugate*** of a quaternion $(a, \vec{v})$ to be the quaternion

$$\overline{(a, \vec{v})} = (a, -\vec{v})$$

An immediate application of quaternion conjugation is finding a formula for the inverse of a quaternion, which ones again ends up working out very similarly to the complex case - We have

$$\begin{aligned} \overline{(a, \vec{v})} \cdot (a, \vec{v}) &= (a, -\vec{v}) \cdot (a, \vec{v}) \\ &= (a^2 - \left\langle -\vec{v}, \vec{v} \right\rangle, a\vec{v} - a\vec{v} + (-\vec{v} \times \vec{v})) \\ &= \left(a^2 + \left\|\vec{v}\right\|^2, \vec{0}\right) \in \mathbb{R} \end{aligned}$$

And thus we can simply divide by this value to get our inverse formula:

Corollary 1.4.7: Inverse Quaternion

Let $(a, \vec{v}) \in \mathbb{H}_{\neq 0}$. Then $(a, \vec{v})$ has a multiplicative inverse, which is given by:

$$(a, \vec{v})^{-1} = \frac{(a, -\vec{v})}{a^2 + \|v\|^2}.$$

Definition 1.4.8. Let $q = (a, \vec{v}) \in \mathbb{H}$. Then

$$|q| := \sqrt{\overline{q} \cdot q} = a^2 + v_1^2 + v_2^2 + v_3^2 = \|q\|_2$$

Note how much this once again resembles both the real case $|x| = \sqrt{x^2} = \|x\|_2$ and the complex case $|z| = \sqrt{\overline{z}z} = \|z\|_2$.

1.5 Quaternions and $\mathbb{R}^3$

Theorem 1.5.1: Quaternions encode rotations

Let $\|\vec{v}\| = 1$ and

$$q = (\cos\varphi, \vec{v}\sin\varphi) \in \mathbb{H}$$

Then $qw\overline{q} \in \mathbb{R}^3$ is the vector obtained by rotating w around the axis $\{\lambda\vec{v} : \lambda \in \mathbb{R}\}$ by the angle 2φ.

Since 2π corresponds to a full rotation, φ and $\varphi + \pi$ describe the same rotation. Therefore, there are two different quaternions q and $-q$ associated with each rotation - in topological terms, we say that the quaternions form a ***double cover*** of the space of all rotations.

This phenomenon is actually closely related to the theory of ***spin*** in quantum physics. All known quantum particles that constitute ordinary matter, including protons, neutrons, electrons, neutrinos, and quarks, are ***spin-$\frac{1}{2}$***, meaning that they exhibit a curious behavior where a rotation of 360° leaves them in a different state than they were before, from which they return to their original state only after being rotated by another 360° around the same axis. This lets us mathematically model particles as quaternions, such that a 360° rotation transforms a particle from a state $q \in \mathbb{H}$ to the "rotationally equivalent" state $-q \in \mathbb{H}$.

Corollary 1.5.2. *For every isometry $F : \mathbb{R}^3 \to \mathbb{R}^3$, there exist $u \in \mathbb{R}^3$ and $q \in \mathbb{H}$ such that we have either*

1. *$F(w) = u + qw\overline{q}$, if F preserves orientation, or*
2. *$F(w) = u + q\overline{w}\overline{q}$, if F doesn't preserve orientation.*

This representation turns out to be a lot more computationally efficient than other representations of the same rotation. Other, more intuitively "trivial" representations of rotation also run into issues such as ***gimbal lock***, where they may get "stuck" in certain orientations.

This makes quaternions surprisingly practical in many applied contexts, including game development and animation[1], robotics, and three-dimensional image processing.

[1]For an introduction to quaternions in game development, I highly recommend this excellent talk by Freya Holmer.

Chapter 2

Linear Maps

2.1 Modules and Vector Spaces

Definition 2.1.1: Modules

Let $(R, +, \cdot)$ be a Ring. Then a (left) ***R-Module*** consists of an abelian Group $(M, +)$ equipped with a ***scalar multiplication*** $. : R \times M \to M$, i.e. scalars are elements of R and "vectors" are elements of M, such that:

1. Ring multiplication associates with scalar multiplication:
$$(r \cdot s).m = r.(s.m)$$
2. Scalar multiplication distributes over module addition:
$$r.(m + n) = r.m + r.n$$
3. Scalar multiplication distributes over scalar addition:
$$(r + s).m = r.m + s.m$$

If R is unital, then we say that M is unital if the identity of ring multiplication is the identity of scalar multiplication:
$$1.m = m$$

A ***right R-module*** is defined similarly, but with a scalar multiplication $. : M \times R \to M$, which is applied from the right instead of from the left.

Exercise 2.1.2. *Show that the distinction between right R-modules and left R-modules is only relevant if R is noncommutative, i.e. that any left R-module M over a commutative ring can be turned into a right R-module by defining $m.r = r.m$ and vice versa.*

To keep this text in line with familiar conventions of vector multiplication in $\mathbb{R}^n$, whenever we talk about a module without explicitly specifiying whether scalar multiplication is applied from the right or from the left, it is implied that it is applied from the left.

Exercise 2.1.3. *Let M be a module over R. Let $m \in M$ and $r \in R$. show that:*

1. $0_R.m = 0_M$
2. $r.0_M = 0_M$
3. $(-r).m = r.(-m) = -(r.m)$

Exercise 2.1.4. *Verify that:*

1. *Every ring R forms a module over itself, where scalar multiplication is given by ring multiplication.*
2. *$\mathbb{R}^n$ forms a module over $\mathbb{R}$, with scalar multiplication defined the usual way. More generally, given any ring R, the cartesian product R^n forms an R-module where scalar multiplication is given by ring multiplication applied componentwise.*
3. *Any abelian group A can be turned into a non-unital module over any ring R by equipping it with the trivial scalar multiplication*

$$r.a = 0_A$$

Definition 2.1.5: Vector Spaces

A module over a field (or skew field) $\Bbbk$ is known as a $\Bbbk$-***vector space***.

Definition 2.1.6: Linear Maps

A ***module homomorphism*** is a structure-preserving map $F : M \to N$ between two modules over the same ring R, i.e. given $m_1, m_2 \in M$ and $r \in R$, we have that:

1. F is ***additive***, i.e. it is compatible with module addition:

$$F(m_1 +_M m_2) = F(m_1) +_N F(m_2)$$

2. F is ***homogenous***, i.e. it is compatible with scalar multiplication:

$$F(r._M m) = r._N F(m)$$

Module homomorphisms are also known as ***linear maps***.

As the name suggests, the main goal of linear algebra is to generalize the concept of a ***linear map***, and to figure out properties and applications of these maps.

Exercise 2.1.7. *Show that any linear map $F : M \to N$ fulfills*

$$F(0_M) = 0_N$$

Exercise 2.1.8. *Show that any linear map $F : R \to R$ from a unital ring into itself can be represented as multiplication by a constant, i.e. $F(r) = a.r = a \cdot_R r$ for some $a \in R$.*

Exercise 2.1.9.

1. *Show that complex conjugation is a linear map $\mathbb{C} \to \mathbb{C}$ if we view $\mathbb{C}$ as a module over $\mathbb{R}$.*
2. *Show that complex conjugation is not linear if we view $\mathbb{C}$ as a module over $\mathbb{C}$.*

2.2 Subspaces and Quotient Spaces

2.3 Bases and Dimension

Definition 2.3.1: Linear Combinations

Let M be an R-module, let $(m_i)_{i \in I} \subseteq M$ and $(r_i)_{i \in I} \subseteq R$. Then a ***linear combination*** is a sum of the form

$$\sum_{i \in I} r_i . m_i$$

Definition 2.3.2: Vector Span

Let M be an R-module, let $S = (m_i)_{i \in I} \subseteq M$. Then the span of S is the set $\langle S \rangle$ of all linear combinations over elements of S, i.e.

$$\langle S \rangle = \left\{ \sum_{i \in J \subseteq I} r_i . m_i \;\middle|\; (r_i)_{i \in I} \subseteq R, m_i \in S \right\}$$

For example, the span of one vector $v \in \mathbb{R}^3$ is the line through the origin on which v lies. The span of two orthogonal vectors $v, w \in \mathbb{R}^3$ is the plane through the origin on which v and w lie.

2.4 Matrices

2.5 Systems of Linear Equations

2.6 The Method of Least Squares

Chapter 3

Affine Spaces

Theorem 3.0.1: Intercept Theorem

Let $v_1, v_2 \in \mathbb{R}$ be linearly independent. Let L_1 be a line through v_2 and v_1. Then if we have $\alpha, \beta \in \mathbb{R}_{\neq 0}$, and L_2 is a parallel line through αv_2 and βv_1, we must have $\alpha = \beta$.

Proof. Since L_1 and L_2 are parallel, their direction vectors must point in the same direction, which means there must be a $\mu \in \mathbb{R}_{\neq 0}$ such that

$$\begin{aligned} \mu(v_2 - v_1) &= \mu(\alpha v_2 - \beta v_1) \\ \implies \mu v_2 - \mu v_1 &= \alpha v_2 - \beta v_1 \\ \implies \mu v_2 - \alpha v_2 &= \mu v_1 - \beta v_1 \\ \implies (\mu - \alpha) v_2 &= (\mu - \beta) v_1 \end{aligned}$$

Since v_2 and v_1 are linearly independent, this implies $\mu - \alpha = \mu - \beta = 0$, i.e. $\alpha = \beta = \mu$. □

Theorem 3.0.2. *Every injective map $\Phi : \mathbb{R}^2 \to \mathbb{R}^2$ mapping 0 to 0 and affine lines to affine lines must be linear.*

Proof. Since Φ is linear with $\Phi(0) = 0$, it must map lines through the origin to lines through the origin. Let

$$\begin{aligned} \Phi(e_1) &:= v_1 \\ \Phi[\lambda e_1] &:= L_1 = \{\lambda v_1\} \\ \Phi(e_2) &:= v_2 \\ \Phi[\lambda e_2] &:= L_2 = \{\lambda v_2\} \end{aligned}$$

Since Φ is injective, L_1 and L_2 can only intersect at $\{0\}$, and thus v_1 and v_2 must be linearly independent.

Now, let's take a closer look at images of individual points. Since

$\Phi[\lambda e_1] = \{\lambda v_1\}$ and $\Phi[\lambda e_2] = \{\lambda v_2\}$, we must have

$$\Phi[\lambda e_1] = \varphi_1(\lambda)v_1$$
$$\Phi[\lambda e_2] = \varphi_2(\lambda)v_2$$

for some $\varphi_1, \varphi_2 : \mathbb{R} \to \mathbb{R}$.

Now, notice that the line through λe_1 and λe_2 is parallel to the line through e_1 and e_2. Since Φ is injective, it must preserve parallelism, and so the line through v_1 and v_2 must be parallel to the line through $\varphi_1(\lambda)v_1$ and $\varphi_2(\lambda)v_2$. The intercept theorem now gives us $\varphi_1(\lambda) = \varphi_2(\lambda) = \mu$. We thus write $\varphi(\lambda) := \varphi_1(\lambda) = \varphi_2(\lambda)$.

Since Φ is injective, φ must be as well, since otherwise we would have some v with $\Phi(\lambda_1 v) = \Phi(\lambda_2 v)$ despite $\lambda_1 v \neq \lambda_2 v$. We have also have $\varphi(0) = 0$ and $\varphi(1) = 1$, since $\Phi(0) = 0$ and $\Phi(e_1) = v_1$.

Now, we want to show that $\Phi(v + w) = \Phi(v) + \Phi(w)$ holds for linearly independent v, w. Since Φ preserves parallelism, it must map parallelograms to parallelograms. Therefore, the parallogram spanned by v and w must be mapped to the parallogram spanned by $\Phi(v)$ and $\Phi(w)$. This means that the upper corner $v + w$ gets mapped to the upper corner $\Phi(v) + \Phi(w)$ which gives us $\Phi(v + w) = \Phi(v) + \Phi(w)$ as desired.

Putting our insights together, we now know that

$$\begin{aligned}\Phi(\lambda_1 e_1 + \lambda_2 e_2) &= \Phi(\lambda_1 e_1) + \Phi(\lambda_2 e_2)\\ &= \varphi(\lambda_1)\Phi(e_1) + \varphi(\lambda_2)\Phi(e_2)\end{aligned}$$

it only remains to be shown that φ is the identity. For this, consider that we have:

$$\begin{aligned}(\varphi(a) + \varphi(b))\Phi(e_1) &= \varphi(a)\Phi(e_1) + \varphi(b)\Phi(e_1)\\ &= \Phi(ae_1 + be_1\\ &= \Phi((a + b)e_1)\\ &= \varphi(a + b)\Phi(e_1),\end{aligned}$$

which gives us $\varphi(a) + \varphi(b) = \varphi(a + b)$. For $a \neq 0$, the line through e_1 and ae_2 must point in the same direction as the line through be_1 and bae_2, so the line through $\Phi(e_1)$ and $\varphi(a)\Phi(e_2)$ must be parallel to the line through $\varphi(b)\Phi(e_1)$ and $\varphi(ab)\Phi(e_2)$, from which the intercept theorem gives us

$$\begin{aligned}&\frac{\varphi(ab)}{\varphi(a)} = \varphi(b)\\ \implies\ &\varphi(ab) = \varphi(a)\varphi(b)\end{aligned}$$

Since we have $\varphi(0) = 0$, $\varphi(1) = 1$, $\varphi(a+b) = \varphi(a)+\varphi(b)$, $\varphi(ab) = \varphi(a)\varphi(b)$, our map $\varphi : \mathbb{R} \to \mathbb{R}$ must be a field homomorphism, of which the identity is the only one. $\square$

Chapter 4

Determinants

4.1 Defining Volume

4.2 Determinants of Endomorphisms

4.3 Orientation

Chapter 5

Eigenvectors and Eigenvalues

Definition 5.0.1: Eigenvalues and Eigenvectors

Let k be a field, $F : V \to V$ an Endomorphism, and $\lambda \in k$. Then the ***λ-eigenspace*** of F is the set

$$\mathrm{Eig}(F, \lambda) := \left\{\vec{v} \in V : F(\vec{v}) = \lambda\vec{v}\right\} = \left\{\vec{v} \in V : (F - \lambda\mathrm{id}_V)(\vec{v}) = 0\right\} = \ker(F - \lambda\mathrm{id}_V)$$

If $F(\vec{v}) = \lambda\vec{v}$, with $\vec{v} \neq \vec{0}$, we say that $\vec{v}$ is an ***eigenvector*** of F with ***eigenvalue*** λ.

Note that we exclude $\vec{0}$ from being an eigenvector, since $F(\vec{0}) = \lambda\vec{0}$ trivially holds for every endomorphism. Thus every λ would be an eigenvalue, which would make the definition pointless. We do, however, allow 0 as an eigenvalue - the 0-eigenspace of F is simply its kernel. Sometimes it ends up being useful to rephrase our definition of eigenspaces in terms of kernels:

$$\begin{aligned}
\mathrm{Eig}(F, \lambda) &= \left\{\vec{v} \in V : F(\vec{v}) = \lambda\vec{v}\right\} \\
&= \left\{\vec{v} \in V : F(\vec{v}) - \lambda\vec{v} = 0\right\} \\
&= \left\{\vec{v} \in V : (F - \lambda\mathrm{id}_V)(\vec{v}) = 0\right\} \\
&= \ker(F - \lambda\mathrm{id}_V)
\end{aligned}$$

This shows that $\mathrm{Eig}(F, \lambda)$ is a subspace of V - by definition, these subspaces must then be the maximal subspaces that are closed under application of F, i.e. $F(\mathrm{Eig}(F, \lambda)) = \mathrm{Eig}(F, \lambda)$.

Exercise 5.0.2. *In an informal, intuition-based way:*

1. *Find the real eigenvalues and eigenspaces of a reflection of* $\mathbb{R}^2$ *along an axis* $\left\{\lambda\vec{v} : \lambda \in \mathbb{R}\right\}$
2. *Find a rotation of* $\mathbb{R}^2$ *that has no real eigenvalues. Find two different rotations of* $\mathbb{R}^2$ *that do have real eigenvalues. Show that any rotation of* $\mathbb{R}^2$ *has complex eigenvalues.*
3. *Find the eigenvectors of the derivative map* $\frac{d}{dx}$

Our characterisation $\mathrm{Eig}(F, \lambda) = \ker(F - \lambda\mathrm{id}_V)$ means that, if F is finite-dimensional, we can find all eigenvectors of F with a <u>given</u> eigenvalue λ by solving $(F - \lambda \mathrm{id}_V)\vec{v} = 0$, which is simply a systen of linear equations. Finding the correct eigenvalues ends up being more tricky - in the section after this one, we will work towards proving the following key result:

Proposition 5.0.3. *The eigenvalues of F are given by the zeroes of the* ***characteristic polynomial*** $\chi_F(\lambda) := \det(F - \lambda \mathrm{id}_V)$.

However, before we do so, we want to familiarize ourselves with some of the more basic properties of eigenvectors and eigenvalues.

One important application is that finding eigenvectors lets us diagonalize endomorphisms, greatly easing computation afterwards:

Theorem 5.0.4. *Let V be a vector space over a field k. Let $B = (b_1, \dots, b_n)$ be a basis of V, and let F be an endomorphism represented in B by a matrix A. Then:*

1. *A basis vector b_i is an eigenvector of F iff. the j-th column of A has the form λe_j.*
2. *F is diagonalizable iff. there exists a basis of V consisting of eigenvectors of F.*

Theorem 5.0.5. *Let V be a vector space over a field k. Let $F : V \longrightarrow V$ be an endomorphism. Then eigenvectors of F with different eigenvalues are linearly independent.*

Corollary 5.0.6. *Let V be an n-dimensional vector space over a field k. Let $F : V \longrightarrow V$ be an endomorphism. Then:*

1. *F has no more than n distinct eigenvalues,*
2. *If $(\lambda_i)_{i \in I}$ are mutually distinct eigenvalues of F, the sum*
$$\sum_{i \in I} \mathrm{Eig}(F, \lambda_i) \subseteq V$$
is direct.
3. *F is diagonalizable if and only if V is a direct sum of eigenspaces of F.*
4. *If F has n distinct eigenvalues, it is diagonalizable.*

Corollary 5.0.7. *Let V, W be n-dimensional vector spaces over a field k. Let $F : V \longrightarrow V$, $G : W \longrightarrow W$ be endomorphisms. Then there exists a unique endomorphism $P : V \longrightarrow W$ such that $P \circ F = G \circ P$ if and only if F and G have the same eigenvalues.*

5.1 Characteristic Polynomials

Theorem 5.1.1: Characterstic Polynomial

Let V be a finite-dimensional vector space over a field $\Bbbk$. Let $F : V \longrightarrow V$ be an endomorphism. Then $\lambda \in \Bbbk$ is an eigenvalue of F if and only if

$$\chi_F(\lambda) = \det(\lambda \cdot \mathrm{id}_V - F) = 0_{\Bbbk}$$

We call χ_F the ***characteristic polynomial of F***.

Proof. We know that $\lambda \in \Bbbk$ is an Eigenvalue of F if and only if

$$\ker(F - \lambda \mathrm{id}_V) \neq \{0_V\},$$

i.e. if and only if $F - \lambda \mathrm{id}_V$ isn't injective.

We also know that $F - \lambda \mathrm{id}_V$ isn't injective if and only if it isn't surjective, i.e. if and only if it isn't invertible, i.e. if and only if

$$\det(\lambda.\mathrm{id}_V - F) = \chi_F(\lambda) = 0_{\Bbbk}.$$

□

Exercise 5.1.2. *Show that every endomorphism $F : V \to V$ of a finite-dimensional complex vector space V has at least one eigenvalue.*

Exercise 5.1.3. *Show that the constant term of a characteristic polynomial $\chi_F(\lambda)$ is given by* $(-1)^n \cdot (\det F)$.

(...)

5.2 Algebras and the Cayley-Hamilton Theorem

Theorem 5.2.1: Evaluation map

Let R be a commutative unital ring, and let A be a commutative unital algebra over R. Then, for every $a \in A$, there exists a unique algebra homomorphism $\mathrm{ev}_a : R[X] \to A$ such that:

1. $\mathrm{ev}_a(r) = 1_A.r$ for all constant polynomials $r \in R[X]$,
2. $\mathrm{ev}_A(X) = X$.

We call ev_a the ***evaluation map of $R[X]$ at a***.

Informally, this means that we can take elements of an algebra A over R and plug them into polynomials that were originally only defined on R, and we won't run into issues. In particular, we can plug matrices into polynomials over an underlying ring! Since ev_a is unique, we will generally just write $P(a) := \mathrm{ev}_a(P)$ from now on.

Theorem 5.2.2: Cayley-Hamilton

Let R be a commutative unital ring. Let $A \in R^{n\times n}$. Then $\chi_A(A) = 0$.

5.3 Minimal Polynomials

Theorem 5.3.1: Minimal polynomial

et F be an endomorphism of a vector space V over a field $\mathbb{k}$. Then there exists a unique polynomial $\mu_F \in \mathbb{k}[X]$ such that:

1. μ_F is either normalized or equal to $0_{\mathbb{k}[X]}$
2. μ_F is a divisor of every $P \in \mathbb{k}[X]$ with $P(F) = 0_{\mathrm{End}_{\mathbb{k}}(V)}$

We call μ_F the ***minimal polynomial of F***.

Chapter 6

Normal Forms of Endomorphisms

6.1 Decomposing Vector Spaces using Polynomials

Exercise 6.1.1. *Let G be an abelian group. Show that G becomes a $\mathbb{Z}$-module when equipped with the operation* $n.g = \underbrace{g + \ldots + g}_{n \text{ times}}$.

Exercise 6.1.2. *Let V be a vector space over a field $\Bbbk$. Show that V becomes an* $\mathrm{End}(V)$*-Module when equipped with the operation $F.v = F(v)$.*

Theorem 6.1.3. *Any $\Bbbk[X]$-module is the same thing as a $\Bbbk$-vector space V equipped with a specified endomorphism F, and vice versa.*

Proof.

1. Let V be a $\Bbbk$-vector space. We know from the previous exercise that V is an $\mathrm{End}(V)$-module. Thus, the polynomial evaluation map $\mathrm{ev}_F : \Bbbk[X] \to \mathrm{End}(V)$ turns V into $\Bbbk$-module via

$$P.v = \mathrm{ev}_F(P)(v) = P(F)(v)$$

2. Now let M be a $\Bbbk[X]$-module. Then M becomes a $\Bbbk$-module, i.e. a $\Bbbk$-vector space, if we restrict scalar multiplication to only accept constant polynomials. Then it can easily be seen that $G(v) = X.v$ is a linear transformation.

Now, note that if we start with a vector space (V, F), then use our construction to get a module M, and then turn it back into a vector space, we will obtain our original F again, since

$$G(v) = X._M v = X(F)(v) = F(v)$$

Conversely, if we start with a module M, then turn it into a vector space (V, F), and then back into a module M, we reobtain our original scalar multiplication, since

$$P._V v = P(G)(v) = P(X._M)(v) = P._M v$$

Therefore, the two directions of our construction are inverse to each other, i.e. we actually have a bijection. □

Thus, given a $\Bbbk[X]$-module M, we can write $M = (V, F)$

Exercise 6.1.4. *Let $M = (V, F)$ and $N = (W, G)$ be $\Bbbk[X]$-Modules. Then a $\Bbbk[X]$-linear map $H : M \to N$ is $\Bbbk$-linear and additionally fulfills $H \circ F = G \circ H$, and any $\Bbbk$-linear map H fulfilling these properties is $\Bbbk[X]$-linear.*

Let $M = (V, F)$ be a $\Bbbk[X]$-Module. Then, by definition, a subspace $U \subseteq V$ is a $\Bbbk[X]$-submodule of M if and only if U is closed under scalar multiplication with elements of $\Bbbk[X]$.

Exercise 6.1.5. *Show that U is closed under scalar multiplication with elements of $\Bbbk[X]$ if and only if $Xu \in U$, which itself holds if and only if $F(u) \in U$ holds for all $u \in U$, i.e. U is **F-invariant**.*

Exercise 6.1.6.

1. *Show that the scalar multiplication map $P. = P(F)$ is linear.*
2. *Conclude that $\ker(P(F))$ and $\operatorname{im}(P(F))$ are submodules of M.*
3. *Conclude that $\ker(P(F))$ and $\operatorname{im}(P(F))$ are F-invariant.*

Theorem 6.1.7: Structure theorem for modules over a Euclidean ring

Let M be a module over a euclidean domain R. Let $a = bc \in R$ such that b and c share no divisors and $a.m = 0$ for all $m \in M$. Then we have:

1. $\ker(b.) = \operatorname{im}(c.)$,
2. $\operatorname{im}(b.) = \ker(c.)$,
3. $M = \ker(b.) \oplus \ker(c.)$.

Proof.

1. Let $c.m \in \operatorname{im}(c.)$ for $m \in M$. Then we have $b.(c.m) = a.m = 0$, i.e. $\operatorname{im}(c.) \subseteq \ker(b.)$. Now, applying the Euclidean algorithm gives us $1 = k_1 b + k_2 c$. If we now let $m \in \ker(b.)$, we get:

$$\begin{aligned} m &= 1m \\ &= (k_1 b + k_2 c)m \\ &= k_1 \underbrace{(bm)}_{=0} + k_2(cm) \\ &= c(k_2 m) \in \operatorname{im}(c.) \end{aligned}$$

   Thus, we also have $\ker(b.) \subseteq \operatorname{im}(c.)$, i.e. $\ker(b.) = \operatorname{im}(c.)$.

2. $\ker(c.) = \operatorname{im}(b.)$ follows from the same argument by swapping the roles of b and c.

3. Now, let $m \in M$. Then, applying our identity from before, we get

$$\begin{aligned} m &= (k_1 b + k_2 c)m \\ &= b(k_1 m) + c(k_2 m) \\ &\in \operatorname{im}(b.) + \operatorname{im}(c.) \\ &= \ker(b.) + \ker(c.), \end{aligned}$$

which gives us $M = \ker(b.) + \ker(c.)$ as desired.

If we have $m \in \ker(b.) \cap \ker(c.)$, we get

$$m = k_1(b(m)) + k_2(c(m)) = 0,$$

so $\ker(b.) \cap \ker(c.) = \{0\}$, so our sum is direct.

□

Corollary 6.1.8. *Let G be an abelian group. Let $n = mk \in \mathbb{Z}$ such that m and k share no divisors and $n \cdot g = 0$ for all $g \in G$. Then we have*

$$G = \ker(m\cdot) \oplus \ker(k\cdot).$$

Corollary 6.1.9: Chinese Remainder Theorem

Corollary 6.1.10. *Let V be a vector space over a field $\Bbbk$ and $F : V \to V$ be an Endomorphism. Let $P = QR \in \Bbbk[X]$ such that Q and R share no divisors and $P.v = P(F)(v) = 0$ for all $v \in V$. Then we have*

$$V = \ker(Q(F)) \oplus \ker(R(F)).$$

Corollary 6.1.11: P-Generalized Eigenspace Decomposition

Let V be a vector space over a field $\Bbbk$ and $F : V \to V$ be an Endomorphism. Let $P \in \Bbbk[X]$ be a normed polynomial and let $P = P_1^{n_1} \cdot \ldots \cdot P_k^{n_k}$ be its prime factorization. Then we have

$$V = \bigoplus_{i=1}^{k} \ker(P_i^{n_i}(F))$$

6.2 The Jordan Normal Form

Corollary 6.2.1: Generalized Eigenspace Decomposition

Let V be a vector space over a field $\mathbb{k}$ and $F : V \longrightarrow V$ be an Endomorphism. Assume that its minimal polynomial μ_F splits into linear factors over $\mathbb{k}$, i.e.

$$\mu_F(x) = \prod_{i=1}^{k}(x - \lambda_i)^{n_i}$$

with $\lambda_i \in \mathbb{k}$. Then we have

$$V = \bigoplus_{i=1}^{k} \ker((F - \lambda_i \mathrm{id}_V)^{n_i})$$

We call $\ker((F - \lambda_i \mathrm{id}_V)^{n_i})$ the ***generalized λ_i-Eigenspace of F*** and denote it $H(F, \lambda_i)$ (from the German "Hauptraum").

Exercise 6.2.2. *Show that $H(F, \lambda_i)$ contains all eigenvectors of F with eigenvalue λ_i.*

Exercise 6.2.3. *Show that $H(F, \lambda_i)$ is an F-invariant subspace of V, i.e. a subspace such that $F(H(F, \lambda_i)) \subseteq H(F, \lambda_i)$*

Exercise 6.2.4. *Let $V = V_1 \oplus V_2$ be a decomposition of V into F-invariant subspaces. Let B_1 be a basis of V_1, and let B_2 be a basis of V_2. Show that:*

1. *$B = B_1 \cup B_2$ is a basis of V*
2. *If the restriction $F|_{V_1}$ of F to V_1 is represented in basis B_1 as a matrix A_1, and the restriction $F|_{V_2}$ of F to V_2 is represented in basis B_2 as a matrix A_2, then F is represented in basis B as*

$$A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

Exercise 6.2.5. *Let $F : V \longrightarrow W$ be a linear map. Let A be a basis of* $\ker F$*, and let B be a tuple of vectors such that $F(B)$ is a basis of* $\operatorname{im} F$*. Then $A \cup B$ is a basis of V.*

We now want to make use of this to construct a Jordan basis B of a nilpotent Endomorphism N:

Corollary 6.2.6. *Let $N : V \longrightarrow V$ be a nilpotent endomorphism. Then there exists a basis B of V such that:*

1. *$B \cup \{0\}$ is F-invariant, i.e. if $b \in B$, then we have $F(b) \in B$ or $F(b) = 0$.*
2. *For every $b \in B$, there exists at most one element $b' \in B$ with $F(b') = b$.*

*We call such a basis a **Jordan basis**.*

> *Proof.* We prove this theorem in three steps: We construct a basis B, then show it is F-invariant, then show that each element of B has at most one preimage in B.

1. We construct our basis by induction over $n = \dim V$.

   If $n = 0$, then the empty set is a basis (since a sum over the empty set equals the additive identity, i.e. the zero vector) which (vacuously) fulfills our desired properties.

   Now, since the kernel of N is non-trivial, we have $\dim \operatorname{im} N < \dim V$. Therefore, by induction, we can assume we already have a Jordan basis A of $\operatorname{im} N$. We split this basis A into two parts:

   (a) $A_0 := \{a \in A \mid N(a) = 0\}$,

   (b) $A_{\neq 0} := \{a \in A \mid N(a) \neq 0\}$.

   A_0 is by definition a set of linearly independent vectors contained in $\ker A$. By the Steinitz exchange lemma, we can add a set C of Linearly independent vectors such that $A_0 \cup C$ is a basis of $\ker A$.

   $A_{\neq}0$ is the set of all A such that $N(A_{\neq 0}) \subseteq A$. We add a set D containing preimages of all the remaining elements of A, which must exist since $A \subseteq \operatorname{im} N$. By construction, we now have $F(A_{\neq 0} \cup D) = A$.

   Putting everything together, $A_0 \cup C$ is a basis of $\ker\ N$, and $F(A_{\neq 0} \cup D) = A$, which is a basis of $\operatorname{im} N$. Therefore, by exercise 6.2.5, $B = A \cup C \cup D$ is a basis of V.

2. Now, B must be F-invariant since A is F-invariant by definition, C is contained in the kernel of F and thus $F(B) = \{0\}$, and D consists of preimages of A, i.e. $F(D) = A \subseteq B$.

3. Finally, the preimages of A in B are contained in either $A_{\neq 0}$ or D, which are disjoint by construction.

□

Corollary 6.2.7. *Let $F : V \to V$ be a matrix whose characteristic polynomial has $n = \dim(V)$ roots. Then there exists a basis B of V such that ${}_BF_B$ is a block diagonal matrix consisting of Jordan blocks.*

Proof. Generalized Eigenspace decomposition 6.2 shows us that V is the direct sum of the generalized Eigenspaces of F. We know by exercise 6.2.3 that the generalized Eigenspaces of F are F-invariant. Therefore, exercise 6.2.4 tells us that if we find a basis B_i for each generalized Eigenspace $H(F, \lambda_i)$ such that $F|_{H(F,\lambda_i)}$ is represented by a matrix J_i in Jordan normal form, and we combine our bases into a basis $B = \bigcup_{i=1}^{n} B_i$ of V as a whole, then the matrix representation of F in basis B will be a block diagonal matrix consisting of these J_i, and thus F will also be in Jordan normal form.

Now, let F_i be the restriction of $(F - \lambda_i \mathrm{id}_V)$ to $H(F, \lambda_i)$. By definition of the generalized Eigenspace, we have

$$H(F, \lambda_i) = \ker((F - \lambda_i \mathrm{id}_V)^{n_i})$$

therefore, by construction, F_i is nilpotent. Therefore, by theorem **??**, there exists a basis B_i such that ${}_{B_i}F_{i\,B_i}$ is in Jordan normal form, with zeroes on the diagonal. But then, since we have

$$F|_{H(F,\lambda_i)} = F_i + \lambda \mathrm{id}_{H(F,\lambda_i)}$$

the matrix representation of $F|_{H(F,\lambda_i)}$ is also in Jordan form, since we simply have to add λ to the diagonal of ${}_{B_i}F_{i\,B_i}$.

Therefore, the restriction to F to any of its generalized Eigenspaces is in Jordan normal form in basis B_i. As discussed at the beginning of this proof, this means F is in Jordan normal form in basis $B = \bigcup_{i=1}^{n} B_i$. □

Exercise 6.2.8. *Let F be an endomorphism, and let J be its Jordan normal form. Then:*

1. *The algebraic multiplicity of each eigenvalue of F corresponds to the number of times that eigenvalue appears on the diagonal of J.*
2. *The geometric multiplicity of any eigenvalue λ of F corresponds to the number of λ-Jordan blocks with that eigenvalue in J.*
3. *The exponent of $(X - \lambda)$ in the minimal polynomial of F corresponds to the size the of largest λ-Jordan block in J.*

6.3 The Weierstrass Normal Form

Chapter 7

Inner Product Spaces

Chapter 8

Tensor Products

Chapter 9

Revisiting Synthetic Geometry

9.1 Incidence Geometries

Definition 9.1.1: Incidence Geometries

An ***incidence geometry*** is a set X equipped with a set $G \subseteq \mathcal{P}(X)$ of subsets of X known as ***lines***, such that:

1. Each line contains at least two points,
2. For every pair of points, there exists exactly one line containing both.

Instead of "x is contained in $g \in G$", we often say "x lies on $g \in G$".

Exercise 9.1.2. *Let R be a division ring. Then $X := R^n$ can be turned into an incidence geometry by defining lines to be all sets of the form*

$$g = \{p + \lambda v : \lambda \in R\},$$

where $p, v \in R^n$.

Moving forward, this will be the canonical choice of incidence geometry on R^n.

Definition 9.1.3. A set of points lying on the same line is called ***colinear***.

Definition 9.1.4. A set of lines containing a common point is called ***confluent***.

Definition 9.1.5. A set of three non-colinear points is known as a ***triangle***.

Definition 9.1.6: Betweenness

A ***betweenness relation*** on an incidence geometry (X, G) is a subset $Z \subseteq X^3$ such that the following properties hold:

1. ***Induced Orderings on Lines:*** On every line $g \in G$, there exists an ordering $\leq$ such that $(x, y, z) \in Z$ if and only if $x \leq y \leq z$ or $x \geq y \geq z$.

2. ***Pasch's Axiom:*** Let $p, q, r \in X$. Let

$$[p, q] := \{x \in X : (p, x, q) \in Z\}$$

denote the ***line segment between p and q***. Then every line $g \in G$ either intersects none of the line segments $[p, q]$, $[q, r]$, $[r, p]$, or at least two.

An incidence geometry equipped with a betweenness relation is known as an ***ordered incidence geometry***.

Exercise 9.1.7. *Let K be an ordered field. Then:*

1. *The set of all tuples (x, y, z), where $x, y, z \in K^n$ fulfill $y = \lambda x + \mu z$ with $\lambda, \mu \in [0, 1] \subset K$ and $\lambda + \mu = 1$, gives a betweenness relation on K^n.*

2. *There exists an affine isomorphism $\varphi : K \cong (g \in G)$ between any line in G and K.*

3. *The ordering on K can be recovered from the betweenness relation on K^n by taking the unique ordering $\leq$ on $g \in G$ that fulfills $\varphi(0) \leq \varphi(1)$.*

Moving forward, this will be the canonical choice of betweenness relation on K^n.

Definition 9.1.8. Given an ordered incidence geometry (X, G, Z), a ***ray*** in X is a set $A \subseteq X$ of the form

$$A = \{x \in g \mid x \leq p\},$$

where $g \in G$ is a line and $p \in g$ is a point on that line.

Definition 9.1.9: Congruence Groups

Given an ordered incidence geometry (X, G, Z), a ***congruence group*** of (X, G, Z) is a group K of automorphism of (X, G, Z) such that every pair of rays in $A, B \subseteq X$ has exactly two elements $h, k \in K$ such that $h(A) = k(A) = B$.

Intuitively, one of these two elements will be a rotation and the other a rotation composed with reflection along an axis.

Exercise 9.1.10. *Let K be an ordered field. Let s be the standard scalar product in K^2. Then the set of s-invariant affine automorphisms of K^2 forms a congruence group of K^2.*

Moving forward, this will be the canonical choice of congruence group on K^2.

Definition 9.1.11. We say that an ordered incidence geometry has the ***least upper bound property*** if every nonempty subset $A \subset g \in G$ that is bounded from above has a least upper bound.

Definition 9.1.12. We call an ordered incidence geometry (X, G, Z) with congruence group K ***near-euclidean*** if (X, G, Z) fulfills the supremum property and G is nonempty.

Definition 9.1.13. We say that an incidence geometry (X, G) fulfills the ***parallel postulate*** iff. for every line $g \in G$ and every point $p \notin G$, there exists at most one line $h \in G$ such that $p \in h$ and $h \cap g = \emptyset$.

Exercise 9.1.14. *Let $K \subsetneq \mathbb{R}$ be a proper subfield of $\mathbb{R}$ containing square roots of every positive element. Then K^2 fulfills the parallel postulate.*

The primary goal of this chapter will be proving the following result:

Theorem 9.1.15: Classification of near-euclidean geometries

Let (X, G, Z) be a near-euclidean incidence geometry. Then:

1. If (X, G, Z) fulfills the parallel postulate, it is isomorphic to $\mathbb{R}^2$.
2. If (X, G, Z) doesn't fulfill the perallel postulate, it is isomorphic to the ***hyperbolic plane*** H.

Chapter 10

Hyperbolic Geometry

10.1 Circle Inversions

Definition 10.1.1. Let E be a Euclidean plane. Then the ***extended Euclidean plane*** is the set

$$\hat{E} = E \cup \{\infty\},$$

adding a "point at infinity" to E.

Definition 10.1.2. We call a subset $L \subseteq \hat{E}$ a ***generalized circle*** if it is either a circle in E or the union of a line in E with $\{\infty\}$. We call the former ***real circles*** and the latter ***extended lines***.

Definition 10.1.3. Let E be a euclidean plane. Then we assign to each generalized circle $L \subseteq \hat{E}$ the ***circle inversion map***

$$s_L : \hat{E} \to \hat{E},$$

which is given by the ordinary reflection at L with the additional rule $s_L(\infty) = \infty$ if L is an extended line, and by

$$c + \vec{v} \mapsto \begin{cases} c + \left(\frac{r}{\|\vec{v}\|}\right)^2 \vec{v} & \vec{v} \neq \vec{0} \\ \infty & \vec{v} = \vec{0} \end{cases}$$

with $\infty \mapsto c$ if L is a circle with center c and radius r.

Exercise 10.1.4. *s_L is an involution, i.e. we have $s_L^2 = \mathrm{id}$.*

Exercise 10.1.5. *For any two points $p, q \in \hat{E}$, there exists a circle inversion s_L with $s_L(p) = q$ and $s_L(q) = p$*

Exercise 10.1.6. *Every circle inversion maps generalized circles to generalized circles.*

10.2 Möbius Transformations

Part IV

Point-Set Topology

Chapter 1

Topological Spaces

A very common axiom is the Hausdorff separation axiom, which says that any two distinct points can be housed orf from each other by disjoint open sets.

"Topology via Logic",
Steven Vickers